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Sem 2
OOP LAB
Tasks

Functions

◆ **LAB Task 1:**

- ✧ **Q1:** Write a program to accept two values from the user and swap them using a third variable.

```
#include <iostream>
using namespace std;

int main() {
    int a, b, temp;

    cout << "Enter two numbers: ";
    cin >> a >> b;

    temp = a;
    a = b;
    b = temp;
    cout << "After swapping: " << a << " " << b << endl;

    return 0;
}
```

Output

Enter two numbers: 4 5
After swapping: 5 4

- ✧ **Q2:** Write a program to accept two values from the user and swap them without using a third variable.

Input

```
#include <iostream>
using namespace std;

int main() {
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    a = a + b;
    b = a - b;
    a = a - b;
    cout << "After swapping: " << a << " " << b << endl;
    return 0;
}
```

Output

Enter two numbers: 9 6
After swapping: 6 9

- ✧ **Q3:** Write a C++ program that get Two Values From User and Perform Mathematical operation using Function.

Input

```
#include <iostream>
using namespace std;
void CalMath(int n1, int n2) {
    cout << "--- Results ---" << endl;
    cout << "Addition: " << n1 + n2 << endl;
    cout << "Subtraction: " << n1 - n2 << endl;
    cout << "Multiplication: " << n1 * n2 << endl;
    if (n2 != 0) {
        cout << "Division: " << (float)n1 / n2 << endl;
    } else {
        cout << "Division: Not possible " << endl;
    }
}
int main() {
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    CalMath(a, b);
    return 0;
}
```

Output

```
Enter two numbers: 4 5
--- Results ---
Addition: 9
Subtraction: -1
Multiplication: 20
Division: 0.8
```

- ✧ **Q4:** Write a Program Swap Two Values without using third variable using Function.

Input

```
#include <iostream>
using namespace std;
void swapNo(int &x, int &y) {
    x = x + y;
    y = x - y;
    x = x - y;
}
int main() {
    int a, b;
    cout << "Enter value for A: ";
    cin >> a;
    cout << "Enter value for B: ";
    cin >> b;
    cout << "Before Swapping: A = " << a << " B = " << b << endl;
    swapNo(a, b);
    cout << "After Swapping: A = " << a << " B = " << b << endl;
}
```

```
    return 0;
}
```

Output

Enter value for A: 7

Enter value for B: 8

Before Swapping: A = 7 B = 8

After Swapping: A = 8 B = 7

✧ **Q5:** Write a Program that get vallue from user and dispay Factorial using Function.

Input

```
#include <iostream>
using namespace std;
long long calculateFactorial(int n) {
    long long fact = 1;
    for(int i = 1; i <= n; i++) {
        fact = fact * i;
    }
    return fact;
}
int main() {
    int number;
    cout << "Enter a positive number: ";
    cin >> number;
    if (number < 0) {
        cout << "Error: Factorial of a negative number does not exist."
<< endl;
    }
    else {
        cout << "The Factorial of " << number << " is: " <<
calculateFactorial(number) << endl;
    }
    return 0;
}
```

Output

Enter a positive number: 9

The Factorial of 9 is: 362880

Arrays

◆ LAB Task 2 :

✧ **Q1:** Write a Program that Declares an arrays and copies its Even Values Into a New Array and Displays it.

Input

```
#include <iostream>
using namespace std;
int main()
{
    int arr[10], z = 0;
    int flag = 0;
    cout << "Enter the values : ";
    for (int i = 0; i < 10; i++)
```

```

{
    cin >> arr[i];
}
for (int j = 0; j < 10; j++)
{
    if (arr[j] % 2 == 0)
        flag++;
}
int arr2[flag];
for (int k = 0; k < 10; k++)
{
    if (arr[k] % 2 == 0)
    {
        arr2[z] = arr[k];
        z++;
    }
}
for (int y = 0; y < flag; y++)
{
    cout << arr2[y];
}
return 0;
}

```

Output

Enter the values : 1

2

3

4

4

5

6

7

9

8

24468